

Datatypes with Loops:

For Loops

1. Strings:

1. Write a program to count the number of vowels in a given string using a for loop.
2. Write a program to reverse a string using a for loop.
3. Write a program to print each character of a string along with its index using a for loop.
4. Write a program to find the frequency of each character in a string using a for loop.
5. Write a program to capitalize the first letter of each word in a string using a for loop.
6. Write a program to replace all vowels in a string with a specific character using a for loop.

2. Lists:

1. Write a program to find the sum of all numbers in a list using a for loop.
2. Write a program to find the largest and smallest numbers in a list using a for loop.
3. Write a program to create a new list with squares of all numbers from an existing list using a for loop.
4. Write a program to count the number of even and odd numbers in a list using a for loop.
5. Write a program to filter all strings from a list containing different data types using a for loop.
6. Write a program to flatten a nested list using a for loop.
(Example: `[[1, 2], [3, 4]]` → `[1, 2, 3, 4]`)

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3. Tuples:

1. Write a program to print each element of a tuple along with its position using a for loop.
2. Write a program to find the count of a specific value in a tuple using a for loop.
3. Write a program to create a new tuple containing only even numbers from an existing tuple using a for loop.
4. Write a program to check if a tuple contains a specific element using a for loop.
5. Write a program to calculate the average of numbers in a tuple using a for loop.
6. Write a program to convert a tuple of strings into a single concatenated string using a for loop.

4. Dictionaries:

1. Write a program to print all keys and their corresponding values in a dictionary using a for loop.
 2. Write a program to calculate the sum of all values in a dictionary using a for loop.
 3. Write a program to create a new dictionary with values doubled from an existing dictionary using a for loop.
 4. Write a program to count the number of keys that start with a specific letter in a dictionary using a for loop.
 5. Write a program to find the length of each value (if the values are strings) in a dictionary using a for loop.
 6. Write a program to merge two dictionaries into one using a for loop.
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While Loops

1. Strings:

1. Write a program to check if a string is a palindrome using a while loop.
2. Write a program to count the number of uppercase letters in a string using a while loop.
3. Write a program to extract and print all digits from a string using a while loop.
4. Write a program to remove all non-alphabetic characters from a string using a while loop.
5. Write a program to count how many times a specific word appears in a sentence using a while loop.
6. Write a program to reverse a given string using while loop.

2. Lists:

1. Write a program to find the average of numbers in a list using a while loop.
2. Write a program to merge two lists alternately into a new list using a while loop.
3. Write a program to remove all occurrences of a specific value from a list using a while loop.
4. Write a program to find the index of the first occurrence of a specific value in a list using a while loop.
5. Write a program to rotate a list to the left by n positions using a while loop.
(Example: $[1, 2, 3, 4, 5] \rightarrow [3, 4, 5, 1, 2]$ for $n = 2$)
6. Write a program to remove duplicate elements from a list using a while loop.

3. Tuples:

1. Write a program to count how many times a specific value appears in a tuple using a while loop.
2. Write a program to calculate the product of all numbers in a tuple using a while loop.
3. Write a program to find the index of a specific element in a tuple using a while loop.
4. Write a program to find the first and last occurrence of a specific value in a tuple using a while loop.
5. Write a program to check if a tuple is sorted in ascending order using a while loop.
6. Write a program to find all unique elements in a tuple using a while loop.

4. Dictionaries:

1. Write a program to find the key associated with the maximum value in a dictionary using a while loop.
2. Write a program to reverse the keys and values in a dictionary using a while loop.
3. Write a program to print all keys from a dictionary where the values are greater than a given threshold using a while loop.
4. Write a program to count how many keys in a dictionary have string values using a while loop.
5. Write a program to remove all key-value pairs from a dictionary where the value is None using a while loop.
6. Write a program to create a new dictionary by swapping the keys and values of an existing dictionary using a while loop.